

15. ANALYSING THE FLOW

In section 13, we constructed a continuous vector field X on the closed surface S . Since in this setting $\mathcal{F}^{cs}$ is in ideal position, its vertical sublamination Λ^{cs} projects via $\pi : M \rightarrow S$ to a geodesic lamination $\Lambda \subset S$.

Together, X and Λ have the following properties:

- (1) the vector field X is tangent to the geodesic lamination Λ ;
- (2) the vector field is zero at a point p if and only if $\pi^{-1}(p)$ is a center circle and we refer to these as the *critical points*;
- (3) each critical point lies on an isolated geodesic in Λ and we call these geodesics the *critical geodesics*;
- (4) each critical geodesic contains exactly one critical point;
- (5) at each critical point, the index of X at p is in $\{-1, 0, +1\}$; moreover, if we consider a small disk D centered at p and split D along the critical geodesic to produce two half-disks, then the index of X at p is in $\{-\frac{1}{2}, 0, \frac{1}{2}\}$ on each of the half-disks.

Item (3) is given by lemma 9.1 and item (5) by lemma 13.1.

The plan now is to first excise the non-isolated leaves of Λ from S , producing a surface with boundary $S_0 \subset S$ where $\Lambda \cap S_0$ consists only of isolated geodesic arcs, each of which is compact. We then split S_0 into a number of “regions” R_i by cutting along the “critical arcs.” By applying the Poincaré–Hopf theorem to the vector field X restricted to a region R_i , we show that at least one of these regions has positive Euler characteristic and therefore must be a topological disk. Its pre-image $\pi^{-1}(R_i)$ must contain half of a vertical leaf of $\mathcal{F}^{cu}$, and this causes a contradiction, completing the overall proof.

For simplicity, we assume that the continuous vector field X integrates to a flow ϕ on S and we explain in the following remark how to adapt the proof when this is not the case.

Remark. The definition of a dumbbell and the proof of proposition 14.1 do not actually rely on X integrating to a flow ϕ on all of S . We can instead define an outflow edge σ as having a small neighborhood U such that any integral curve starting at a point $x \in U$ and tangent to X must remain in U until it hits σ . The proof of proposition 14.1 only considers the flow ϕ on the lamination Γ . In our setting, Γ will be a geodesic lamination and so the flow is well defined. After applying the proposition and restricting to a subset $S_0 \subset S$, we may replace X by a smooth approximation such that X is unchanged on the finitely many arcs of $\Lambda \cap S_0$ which contain critical points. Thus, we may freely assume that X integrates to a flow ϕ .

Let Γ be the sublamination of Λ consisting of all of the non-isolated leaves in Λ . Apply proposition 14.1 to Γ and let S_0 denote the subset $S \setminus \text{int}(K)$ given by the proposition. Each boundary component of S_0 is therefore a dumbbell.